

# **R&D DOCUMENT ON POINT-TO-SITE SETUP ON AZURE**

## PREPARED BY :

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## PURPOSE :

To research and demonstrate the step-by-step setup process of a Point-to-Site (P2S) VPN connection in Azure using certificate authentication.

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## **INTRODUCTION: POINT-TO-SITE VPN**

- A point-to-site VPN is a connection between a single client machine and an Azure Virtual Network (VNet).
- It is ideal for remote users, developers, or testers.
- Requires no physical VPN device.
- Secure communication happens via certificates or Azure AD authentication.

## **CERTIFICATION SETUP FOR AUTHENTICATION**

Point-to-Site VPN in Azure supports certificate-based authentication, which uses a trusted root certificate and client certificates generated from it. This ensures only authorized users can establish VPN connections.

### **Required Certificates:**

- Root Certificate: The trusted certificate that is uploaded to Azure VPN Gateway and used to validate client certs.
- Client Certificate: Installed on each client machine. Proves identity to Azure VPN Gateway.

## **STEP-BY-STEP SETUP**

## ① Create a Virtual Network

- Go to Azure Portal → Search “Virtual networks”
- Click Create
- Provide:
  - Name, Region
  - Address Space
  - Subnet
- Click Review + Create

The screenshot shows the 'Create virtual network' page in the Microsoft Azure portal. The page has a blue header with the Microsoft Azure logo and a search bar. Below the header, there's a breadcrumb trail: Home > Create a resource > Virtual network >. The main heading is 'Create virtual network'. Below this, there are tabs: Basics, IP Addresses (selected), Security, Tags, and Review + create. The 'IP Addresses' tab contains the following information:

The virtual network's address space, specified as one or more address prefixes in CIDR notation (e.g. 192.168.1.0/24).

IPv4 address space

10.0.0.0/16 10.0.0.0 - 10.0.255.255 (65536 addresses)

☐ Add IPv6 address space ⓘ

The subnet's address range in CIDR notation (e.g. 192.168.1.0/24). It must be contained by the address space of the virtual network.

+ Add subnet ⓘ Remove subnet

| Subnet name                      | Subnet address range | NAT gateway |
|----------------------------------|----------------------|-------------|
| <input type="checkbox"/> default | 10.0.0.0/24          | -           |

ⓘ Use of a NAT gateway is recommended for outbound internet access from a subnet. You can deploy a NAT gateway and assign it to a subnet after you create the virtual network. [Learn more](#) ⓘ

At the bottom, there are three buttons: 'Review + create' (blue), '< Previous' (grey), and 'Next : Security >' (grey). There is also a link 'Download a template for automation'.

## ② Create a Virtual Network Gateway

- Go to Azure Portal → Search "Virtual Network Gateways"
- Click Create
- Fill details:

- Name
- Region
- Gateway Type
- VPN Type
- SKU
- Virtual Network
- Public IP address

- Click Review + Create

## Create virtual network gateway ...

**Basics**   Tags   Review + create

Azure has provided a planning and design guide to help you configure the various VPN gateway options. [Learn more](#) ↗

### Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources. ↗

Subscription \*  ▼

Resource group ⓘ TestRG1 (derived from virtual network's resource group)

### Instance details

Name \*  ✓

Region \*  ▼

[Deploy to an edge zone](#) ↗

Gateway type \* ⓘ ☒ VPN ☐ ExpressRoute

SKU \* ⓘ  ▼

Generation ⓘ  ▼

Virtual network \* ⓘ  ▼

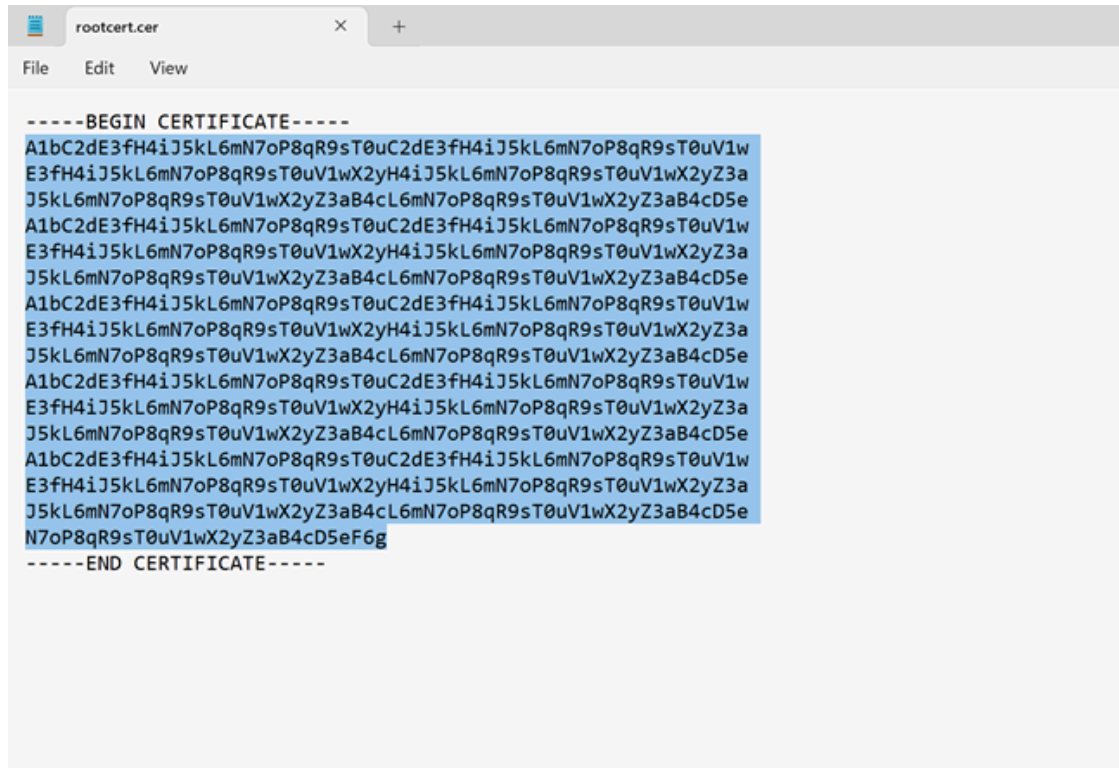
[Create virtual network](#)

Subnet ⓘ  ▼

ⓘ Only virtual networks in the currently selected subscription and region are listed.

## ③ Generate Certificates for Authentication

- a. Create a Root Certificate
- b. Export Public Key
- c. Create a Client Certificate



#### 4) Configure Point-to-Site on the VPN Gateway

- Go to your VPN Gateway → Click Point-to-site configuration
- Click Configure now
- Fill:
  - Address Pool
  - Tunnel Type
  - Authentication Type
  - Root Certificate
- Click Save

**VNet1GW | Point-to-site configuration** ☆ ...  
Virtual network gateway

Search × << Save Discard Delete Download VPN client

Dynamic Static

Root certificates

| Name                 | Public certificate data       |
|----------------------|-------------------------------|
| P2SRootCertificate ✓ | A1bC2dE3fH4iJ5kL6mN7oP8q... ✓ |
|                      |                               |

Revoked certificates

| Name | Thumbprint |
|------|------------|
|      |            |

Additional routes to advertise

Overview  
Activity log  
Access control (IAM)  
Tags  
Diagnose and solve problems  
Settings  
Configuration  
Connections  
**Point-to-site configuration**  
NAT Rules  
Maintenance

## 5 Download VPN Client

- Go to the Point-to-Site configuration tab
- Click Download VPN client
- Choose the correct format based on your client OS (Windows, macOS, Linux)
- Extract the package and run the installer on the client machine

**VNet1GW | Point-to-site configuration** ☆ ...  
Virtual network gateway

Search (Ctrl+/) << Save Discard Delete Download VPN client

Overview  
Activity log  
● Access control (IAM)

Address pool \*

172.16.201.0/24 ✓

## 6 Connect to Azure via VPN

- On the client machine:
  - Go to Settings → Network & Internet → VPN
  - Find the installed VPN profile
  - Click Connect
  - You should now be securely connected to your Azure VNet

## Create connection ...

Basics Settings Tags Review + create

Create a secure connection to your virtual network by using VPN Gateway or ExpressRoute.

[Learn more about VPN Gateway](#)

[Learn more about ExpressRoute](#)

### Project details

|                  |                     |
|------------------|---------------------|
| Subscription *   | Content Development |
| Resource group * | TestRG1             |

[Create new](#)

### Instance details

|                     |                      |
|---------------------|----------------------|
| Connection type * ⓘ | Site-to-site (IPsec) |
| Name *              | VNet1toSite1         |
| Region *            | East US              |

Review + create

Previous

Next : Settings >

[Download a template for automation](#)

## VERIFYING CONNECTION

- Use ipconfig to verify the IP address from the VPN address pool.
- Use ping <VM-IP> or Remote Desktop to access resources in the VNet.
- Use Network Watcher or Connection Troubleshoot to debug issues.



### TOOLS USED

| Tool         | Purpose                  |
|--------------|--------------------------|
| Azure Portal | GUI to manage resources. |

|                 |                                     |
|-----------------|-------------------------------------|
| PowerShell      | To generate and manage certificates |
| VPN Client      | To connect from client machine      |
| Network Watcher | To monitor and troubleshoot VPN     |

## Conclusion

Setting up a Point-to-Site VPN using certificate authentication is a secure and scalable solution for enabling remote users to access Azure resources without deploying dedicated VPN hardware.

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